Gradient Formulas

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1 Gradient Formulas

The likelihood for a point x_i is

$$p(x_i \mid \pi, \mu, \sigma^2) = \pi_1 \mathcal{N}(x_i \mid \mu_1, \sigma_1^2) + \pi_2 \mathcal{N}(x_i \mid \mu_2, \sigma_2^2).$$

The log-likelihood for m data points is

$$\mathcal{L} = \sum_{i=1}^{m} \log \left(\sum_{k=1}^{2} \pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k^2) \right).$$

Define the responsibility of component k for data point i:

$$\gamma_{ik} = \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)}{\sum_{j=1}^{2} \pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)}.$$

We also know:

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2} \right).$$

1.1 Mean Gradient

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \mu_k} &= \sum_{i=1}^{m} \frac{1}{\sum_{j=1}^{2} \pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)} \cdot \frac{\pi_k}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x_i-\mu_k)^2}{2\sigma_k^2} \right) \cdot \frac{x_i - \mu_k}{\sigma_k^2} && (1) \\
&= \sum_{i=1}^{m} \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)}{\sum_{j=1}^{2} \pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)} \cdot \frac{x_i - \mu_k}{\sigma_k^2} && (2) \\
&= \sum_{i=1}^{m} \gamma_{ik} \cdot \frac{x_i - \mu_k}{\sigma_k^2}. && (3)
\end{aligned}$$

1.2 Variance Gradient

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \sigma_k^2} &= \sum_{i=1}^{m} \frac{1}{\sum_{j=1}^{2} \pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)} \cdot \pi_k \cdot \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x_i-\mu_k)^2}{2\sigma_k^2} \right) \cdot \left(-\frac{1}{2\sigma_k^2} + \frac{(x_i-\mu_k)^2}{2\sigma_k^4} \right) && (4) \\
\frac{\partial \mathcal{L}}{\partial \sigma_k^2} &= \sum_{i=1}^{m} \frac{1}{\sum_{j=1}^{2} \pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)} \cdot \pi_k \cdot \mathcal{N}(x_i \mid \mu_k, \sigma_k^2) \cdot \left(-\frac{1}{2\sigma_k^2} + \frac{(x_i-\mu_k)^2}{2\sigma_k^4} \right) && (5) \\
&= \frac{1}{2} \sum_{i=1}^{m} \gamma_{ik} \left(\frac{(x_i-\mu_k)^2}{\sigma_k^4} - \frac{1}{\sigma_k^2} \right) && (6)
\end{aligned}$$

1.3 Mixture weights gradient (with constraint)

$$\frac{\partial \mathcal{L}}{\partial \pi_k} = \sum_{i=1}^{m} \frac{1}{\sum_{j=1}^{2} \pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)} \cdot \mathcal{N}(x_i \mid \mu_k, \sigma_k^2) \tag{7}$$

$$= \sum_{i=1}^{m} \frac{\pi_k \mathcal{N}(x_i \mid \mu_k, \sigma_k^2)}{\sum_{j=1}^{2} \pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)} \cdot \frac{1}{\pi_k} \tag{8}$$

$$= \sum_{i=1}^{m} \frac{\gamma_{ik}}{\pi_k} \tag{9}$$